晶采光電科技股份有限公司
AMPIRE CO., LTD.

SPECIFICATIONS FOR LCD MODULE

CUSTOMER	
CUSTOMER PART NO.	
AMPIRE PART NO.	AM-800480SETMQW-TW2H
APPROVED BY	
DATE	

☑ Approved For Specifications
☐Approved For Specifications & Sample

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RECORD OF REVISION

Revision Date	Page	Contents	Editor
2014/04/21	--	New Release	Rober

1. INTRODUCTION

Ampire Display Module is a color active matrix TFT-LCD that uses amorphous silicon TFT as a switching device. This model is composed of a TFT-LCD panel, driving circuit and touch panel. This TFT-LCD has a high resolution (800(R.G.B) X 480) and can display up to 262,144 colors.

1-1. Features

- 7” WVGA (16:9 diagonal) configuration
- Input interface voltage : 3.3V
- Interface: 40 pin pitch 0.5 FFC
- Capacitive touch panel VDD : 5V

1-2. LCD Controller SSD1963 Feature:

(1) MCU interface: i80/M68 series MCU interface (default: i80 series).
(2) Pixel data format: 8, 9, 16 and 18 bit.
(3) Arbitrary display memory starts position selection.

1-3. Touch panel Feature:

(1) Capacitive touch panel with 2 fingers input
(2) I^2C interface for touch panel controller

2. PHYSICAL SPECIFICATIONS

Item	Specifications	unit
Display resolution(dot)	800RGB (W) x 480(H)	dots
Active area	152.4 (W) x 91.44 (H)	mm
Pixel pitch	0.1905 (W) x 0.1905 (H)	mm
Color configuration	R.G.B Vertical stripe	
Overall dimension	165.0(W) x 104.44(H) x 10.51 max(T)	mm
Brightness	300	cd/m^2
Contrast ratio	400 : 1	
Backlight unit	LED	
Display color	262,144	colors

3. ABSOLUTE MAX. RATINGS

Item	Symbol	Min.	Max.	Unit
LCD Power voltage	VDD	-0.3	4.6	V
LED Power voltage	VLED	0	6.0	V
Operation Temperature	Top	-20	70	℃
Storage Temperature	Tstg	-30	80	℃

4. ELECTRICAL CHARACTERISTICS

4-1 DC Electrical characteristic of the LCD

Item		Symbol	Min.	Typ.	Max.	Unit	Remark
Power supply		VDD	3.0	3.3	3.6	V	
Input Voltage for logic	H Level	V_{IH}	0.7 VDD	--	VDD	V	Note 1
	L Level	V_{IL}	VSS	--	0.3 VDD	V	
Power Supply current (Without B/L)		IDD	-	200	-	mA	Note 2

Note 1: /CS,/WR,/RD,RS,DB0~DB17

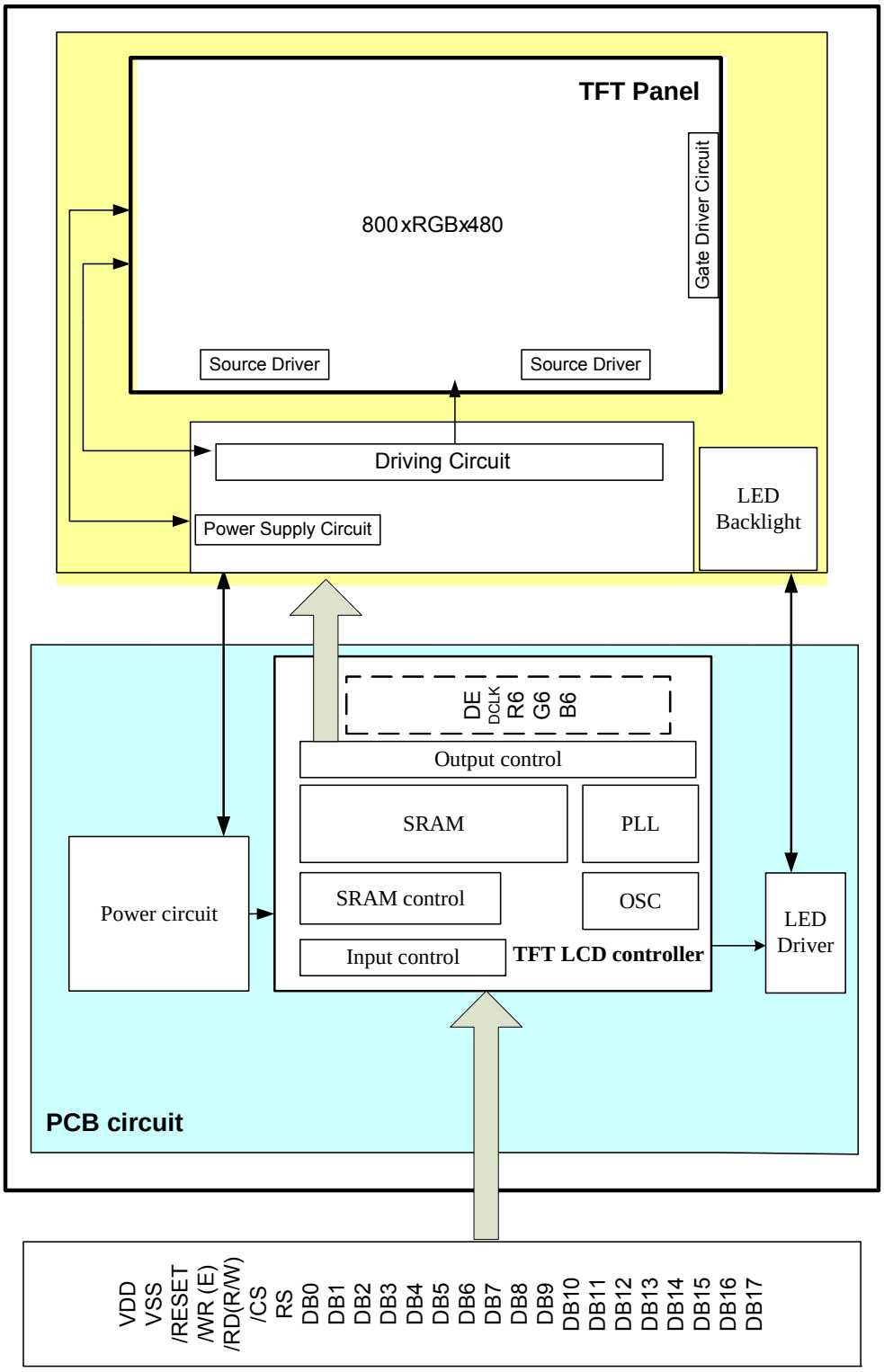
Note 2: fV =60Hz , Ta=25℃ , Display pattern : All Black

4-2 Electrical characteristic of LED Back-light

Ta=25℃

Paramenter	Symbol	Min.	Typ.	Max.	Unit	Condiction
LED driver voltage	VLED	4.5	5	5.5	V	
LED driver current	I_{VLED}	--	400	--	mA	VLED=5V

4-3 Block diagram



5. INTERFACE

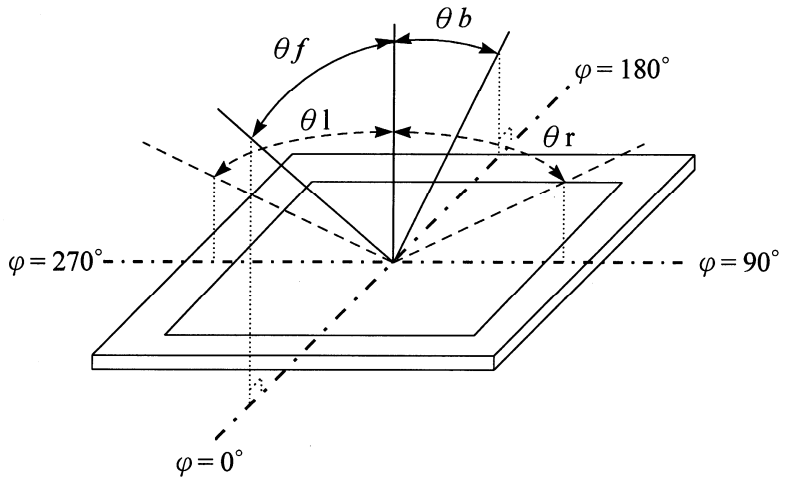
Pin no	Symbol	I/O	Description	Remark
1	DGND	-	GND	
2				
3	VLED	I	LED Power input (5V)	
4	NC	-	No connection	
5	/RESET	I	Reset signal for TFT LCD controller.	
6	RS	I	Register and Data select for TFT LCD controller.	
7	/CS	I	Chip select low active signal for TFT LCD controller.	
8	/WR	I	80mode: /WR low active signal for TFT LCD controller. 68mode: E signal latch on rising edge.	
9	/RD	I	80mode: /RD low active signal for TFT LCD controller. 68mode: R/W signal Hi: read, Lo: write.	
10	DB0	I	Data bus.	
11	DB1	I		
12	DB2	I		
13	DB3	I		
14	DB4	I		
15	DB5	I		
16	DB6	I		
17	DB7	I		
18	DB8	I		
19	DB9	I		
20	DB10	I		
21	DB11	I		
22	DB12	I		
23	DB13	I		
24	DB14	I		
25	DB15	I		
26	DB16	I		
27	DB17	I		
28	NC	-	No connection.	
29	DGND	-	GND	
30	NC	-	No connection.	
31	NC	-	No connection.	
32	NC	-	No connection.	
33	NC	-	No connection.	
34	NC	-	No connection.	
35-37	VDD	-	Power supply for the logic (3.3V).	
38-40	DGND	-	GND.	

6. OPTICAL CHARACTERISTICS

Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Viewing Angle	Front	θ f	CR≧10	50	60	--	deg.	(1)(2)(3)
	Back	θ b		60	70	--		
	Left	θ l		60	70	--		
	Right	θ r		60	70	--		
Contrast ratio		CR	Θ=Φ=0°	250	400	--	--	(1)(3)
Response Time		T_r	Θ=Φ=0°	--	5	10	ms	(1)(4)
		T_f		--	11	16	ms	(1)(4)
Color chromaticity	White	Wx		0.249	0.299	0.349	--	(1)
		Wy		0.278	0.328	0.378		
Luminance		L	Θ=Φ=0°	240	300	--	cd/m²	(1)(5)
Luminance Uniformity		ΔL	Θ=Φ=0°	70	--	--	%	(1)(5)(6)

Note 1: Ta=25°C. To be measured on the center area of panel after 10 minutes operation.

Note 2: Definition of Viewing Angle



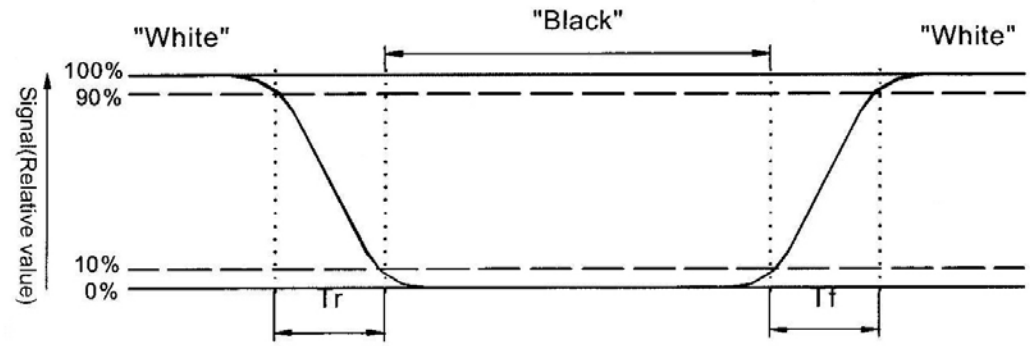
Note 3: Definition of contrast ratio:

Contrast ratio is calculated with the following formula.

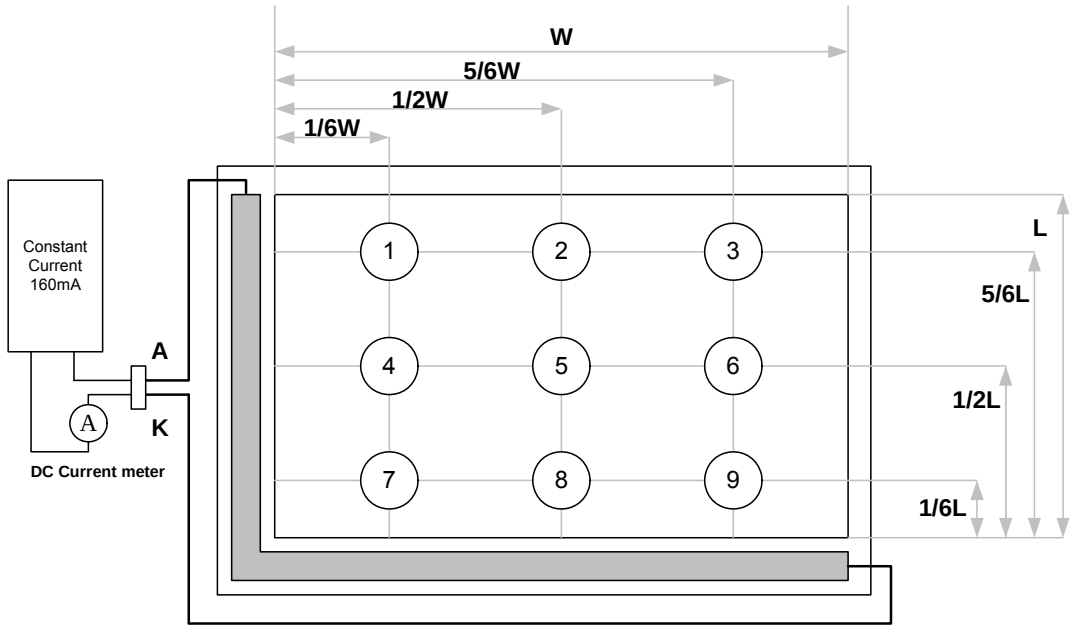
$$\text{Contrast ratio(CR)} = \frac{\text{Photo detector output when LCD is at ”White” state}}{\text{Photo detector Output when LCD is at “Black” state}}$$

Note 4: Definition of response time:

The output signals of photo detector are measured when the input signals are changed from “black“ to “white”(falling time) and from ”white” to “black” (rising time) respectively. The response time is defined as the time interval between the 10% and 90% of amplitudes. Refer to figure as below.



Note 5 : Luminance is measured at point 5 of the display.



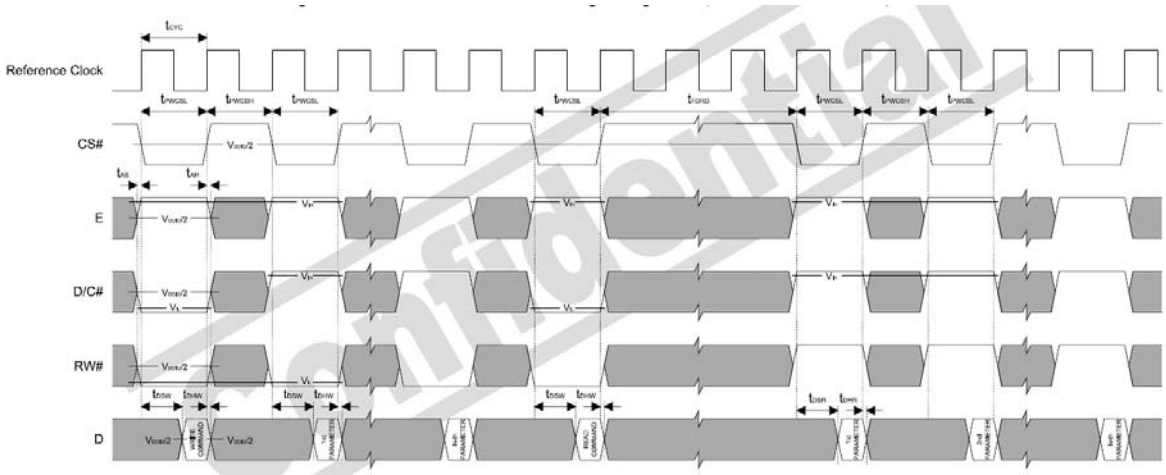
Note 6 : Definition of Luminance Uniformity

ΔL = [L(min.) of 9 points / L(max.) of 9 points] X 100%

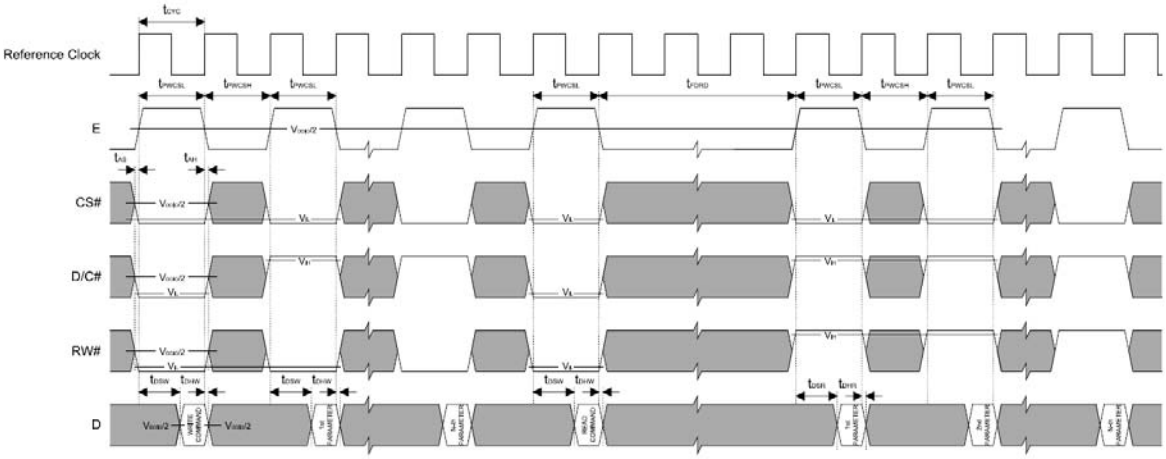
7 LCD Interface Protocol

7.1 M68 Series

Symbol	Parameter	Min	Typ	Max	Unit
t_{CYC}	Reference Clock Cycle Time	9	-	-	ns
t_{PWCSL}	Pulse width CS# or E low	1	-	-	t_{CYC}
t_{PWCSH}	Pulse width CS# or E high	1	-	-	t_{CYC}
t_{FDRD}	First Data Read Delay	5	-	-	t_{CYC}
t_{AS}	Address Setup Time	1	-	-	ns
t_{AH}	Address Hold Time	1	-	-	ns
t_{DSW}	Data Setup Time	4	-	-	ns
t_{DHW}	Data Hold Time	1	-	-	ns
t_{DSR}	Data Access Time	-	-	5	ns
t_{DHR}	Output Hold time	1	-	-	ns



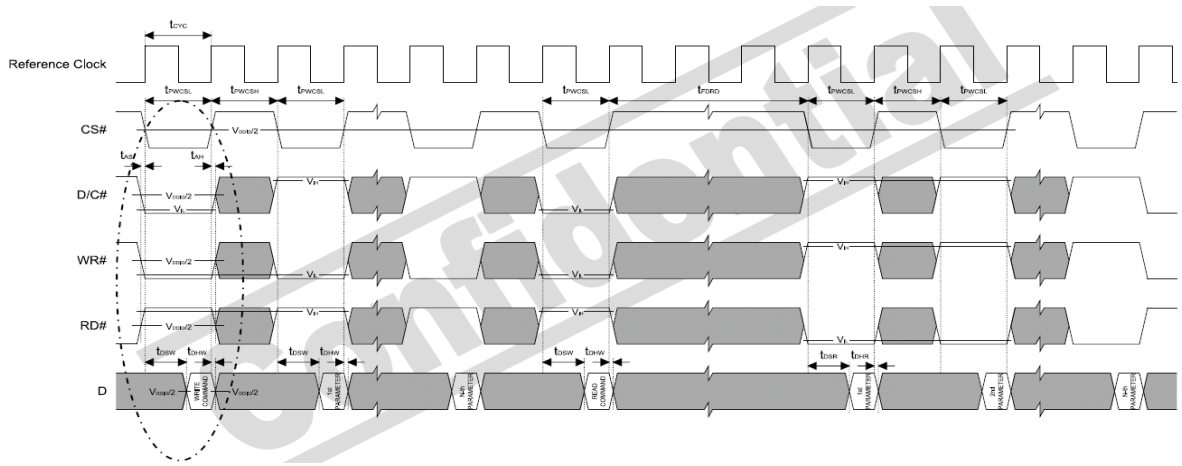
6800 Mode Timing Diagram (Use CS# as Clock)

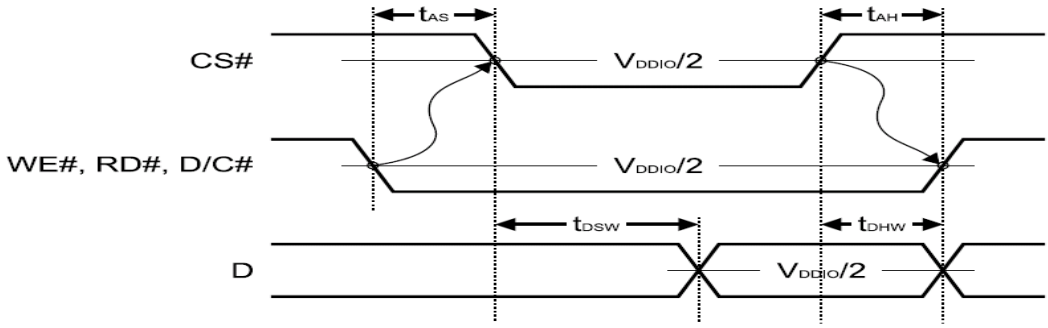


6800 Mode Timing Diagram (Use E as Clock)

7.2 i80 Series

Symbol	Parameter	Min	Typ	Max	Unit
t_{CYC}	Reference Clock Cycle Time	9	-	-	ns
t_{PWCSL}	Pulse width CS# low	1	-	-	t_{CYC}
t_{PWCSH}	Pulse width CS# high	1	-	-	t_{CYC}
t_{FDRD}	First Read Data Delay	5	-	-	t_{CYC}
t_{AS}	Address Setup Time	1	-	-	ns
t_{AH}	Address Hold Time	1	-	-	ns
t_{DSW}	Data Setup Time	4	-	-	ns
t_{DHW}	Data Hold Time	1	-	-	ns
t_{DSR}	Data Access Time	-	-	5	ns
t_{DHR}	Output Hold time	1	-	-	ns





7.3 Data transfer order Setting

Interface	Cycle	D[23]	D[22]	D[21]	D[20]	D[19]	D[18]	D[17]	D[16]	D[15]	D[14]	D[13]	D[12]	D[11]	D[10]	D[9]	D[8]	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
24 bits	1st	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
18 bits	1st							R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
16 bits (565 format)	1st									R5	R4	R3	R2	R1	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1
16 bits	1st									R5	R4	R3	R2	R1	R0	X	X	G5	G4	G3	G2	G1	G0	X	X
	2nd									B5	B4	B3	B2	B1	B0	X	X	R5	R4	R3	R2	R1	R0	X	X
	3rd									G5	G4	G3	G2	G1	G0	X	X	B5	B4	B3	B2	B1	B0	X	X
9 bits	1st																R5	R4	R3	R2	R1	R0	G5	G4	G3
	2nd																G2	G1	G0	B5	B4	B3	B2	B1	B0
8 bits	1st																	R5	R4	R3	R2	R1	R0	X	X
	2nd																	G5	G4	G3	G2	G1	G0	X	X
	3rd																	B5	B4	B3	B2	B1	B0	X	X

X: Don't Care

8. Touch Panel SPECIFICATION

8.1 Basic Characteristic

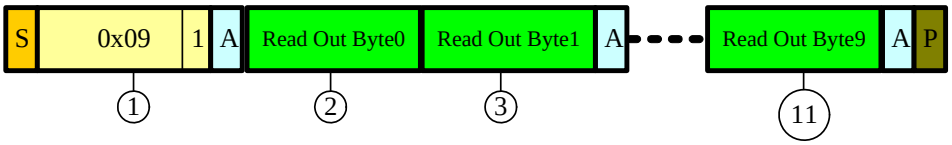
ITEM	SPECIFICATION
Type	Projective Capacitive Touch Panel
Activation	Two-fingers or Signal-finger
X/Y Position Reporting	Absolute Position
Touch Force	No contact pressure required
Calibration	No need for calibration
Report Rate	Approx 25 points/sec
Control IC	EETI EXC7200 +EX5404

8.2 Interface

CN6		
Pin No.	**Symbol**	**Function**
1	GND	POWER GND
2	SDA	I2C DATA
3	SCL	I2C CLOCK
4	VDD	5V
5	INT	Interrupt Request pin
6	RST	Reset pin to Master Chip

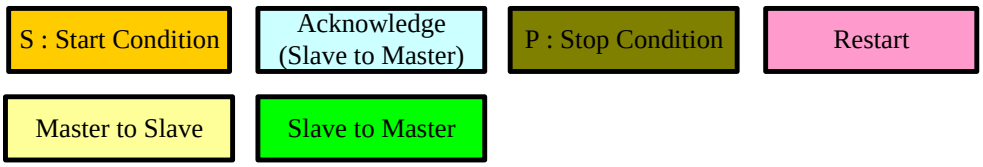
8.3 7-bit I2C address = 0x04.

The complete I2C data format:

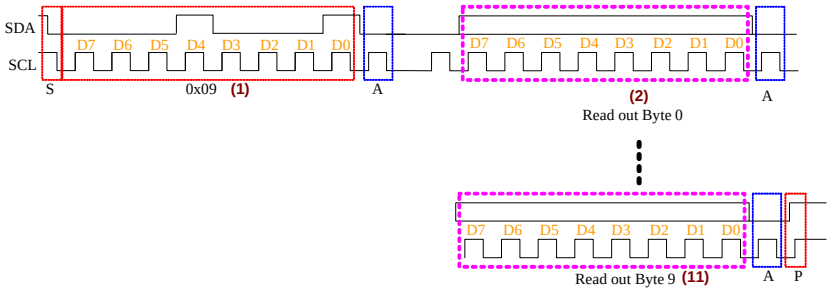


① // 7bit I2C address =0x04 Send I2C Slave address 0x04 Bit1=1 for:write

② ~ ⑪ // Read out Byte0~byte10



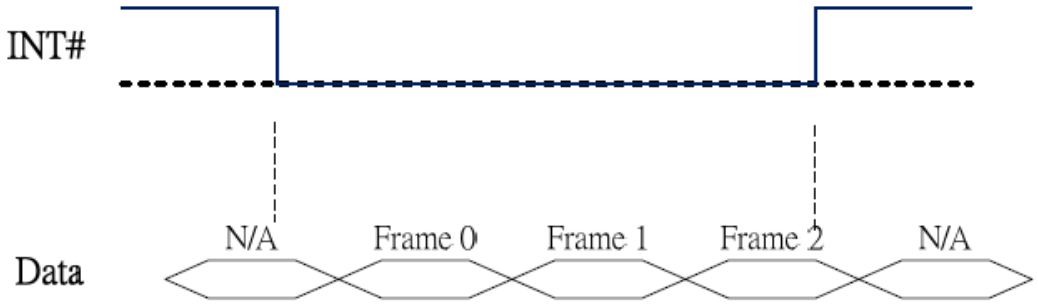
The detail Timing



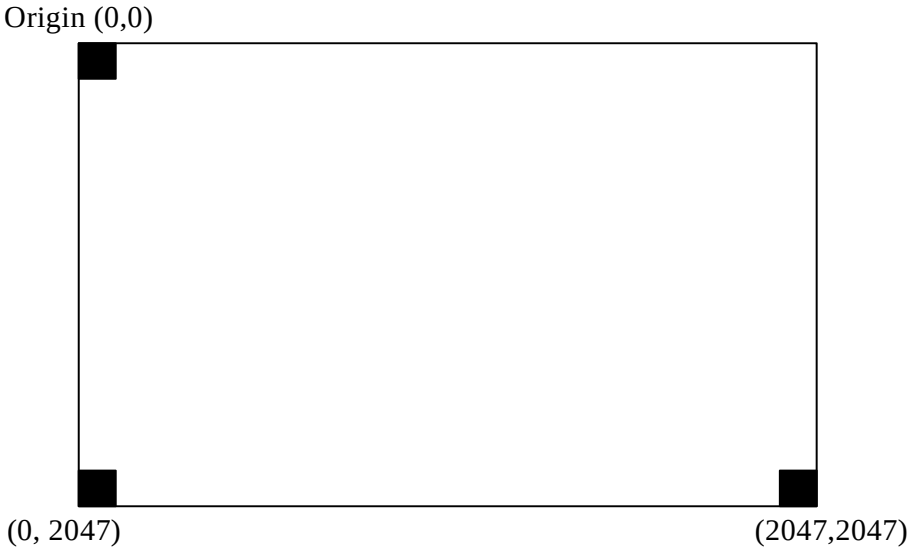
Read Out Byte	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
BYTE0	0	0	0	0	0	1	0	0
	Report ID = 0x04							
BYTE1								
	BIT[7] : When this bit =1 is Valid touch. BIT[6:2] : Contact ID. BIT[1] : In range bit , this bit is always 1. BIT[0] : This bit 1 for touch down , 0 for touch lift Example: 0x83: 1st Touch Down. 0x82: 1st Touch Lift. 0x87: 2nd Touch Down.							

	0x86: 2nd Touch Lift.							
BYTE2	Touch X [3:0]				Don’t care			
	X Coordination Bit [3:0] in BYTE2 Bit [7: 4]							
BYTE3	Touch X [15:4]							
	X Coordination Bit [15:4] in BYTE3							
BYTE4	Touch Y [3:0]				Don’t care			
	Y Coordination Bit [3:0] in BYTE2 Bit [7: 4]							
BYTE5	Touch Y [15:4]							
	Y Coordination Bit [15:4] in BYTE3							
BYTE6								
	Reserved							
BYTE7								
	Reserved							
BYTE8								
	Reserved							
BYTE9								
	Reserved							

Interrupt starts at touch-down event and ends at touch-lift event.
During the period, coordinate report rate is related to the rate which host issues read-coordinate command.



Coordinate



Sample Code :

```
/*************************************************************************
* Function Name    : u8 EXC7200_I2C_CoordRead(u8 Slave_Addr,u8 read_Nbyte , u16    *ByteReturn)
* Description       : Use GPIO Read N byte data from Slave device (Addr) to Host
* Input                : u8 Slave_Addr , Ex:0x04
*                        : u8 read_Nbyte
*                        : Point for touch information
*                            ByteReturn [0] = Point X1
*                            ByteReturn [1] = Point Y1
*                            ByteReturn [2] = T.B.D
*                            ByteReturn [3] = Point Y2
*                            ByteReturn [4] = Point Y2
*                            ByteReturn [5] = T.B.D
* Return
*
* #define EXC7200_I2C_CoordRead_1stTouch_Down    0x01
* #define EXC7200_I2C_CoordRead_1stTouch_Lift    0x81
* #define EXC7200_I2C_CoordRead_2ndTouch_Down    0x02
* #define EXC7200_I2C_CoordRead_2ndtTouch_Lift    0x82
*************************************************************************/
#define EXC7200_I2C_CoordRead_1stTouch_Down    0x01
#define EXC7200_I2C_CoordRead_1stTouch_Lift    0x81
#define EXC7200_I2C_CoordRead_2ndTouch_Down    0x02
#define EXC7200_I2C_CoordRead_2ndtTouch_Lift    0x82
#define EXC7200_I2C_CoordRead_error    0x00

u8 EXC7200_I2C_CoordRead(u8 Slave_Addr,u8 read_Nbyte , u16    *ByteReturn )
{
   u8 *pBuffer ,i ;
   u8 Byte[10] ;


   IO_I2C_start();    // Start Condiction
   IO_I2C_reg_cmd_para((Slave_Addr<<1)+1); // Send I2C Slave address+1 Bit0=1 for:read
                pBuffer=IO_I2C_read_Nbyte(read_Nbyte);      // read 10 byte

                for(i=0;i<10;i++)
                    {
                            Byte[i]=*pBuffer;
                            pBuffer++;
                    }
                if( (Byte[1]==0x83) | (Byte[1]==0x82))    //
                {
                    ByteReturn[0]=((u16)((Byte[3]&0x00ff)<<4))+((u16)((Byte[2]&0x00f0)>>4)); //Point X1
                     ByteReturn[1]=((u16)((Byte[5]&0x00ff)<<4))+((u16)((Byte[4]&0x00f0)>>4)); //Point Y1
                     ByteReturn[2]= 0xFFFF;
                     ByteReturn[3]= 0xFFFF;
                     ByteReturn[4]= 0xFFFF;
                     ByteReturn[5]= 0xFFFF;
                     Previous_X1=ByteReturn[0];
                     Previous_Y1=ByteReturn[1];

                     if ( (Byte[1]==0x83))
                     {
                            return EXC7200_I2C_CoordRead_1stTouch_Down;
                     }
                     if ( (Byte[1]==0x82))
                     {
                            return EXC7200_I2C_CoordRead_1stTouch_Lift;
                     }

   }
```

```
                if( (Byte[1]==0x87) | (Byte[1]==0x86))    //
                {
                    ByteReturn[3]=((u16)((Byte[3]&0x00ff)<<4))+((u16)((Byte[2]&0x00f0)>>4)); //Point X1
                     ByteReturn[4]=((u16)((Byte[5]&0x00ff)<<4))+((u16)((Byte[4]&0x00f0)>>4)); //Point Y1
                     ByteReturn[5]= 0xFFFF;
                     ByteReturn[0]= Previous_X1;
                     ByteReturn[1]= Previous_Y1;
                     ByteReturn[2]= Previous_Z1;

                     if ( (Byte[1]==0x87))
                     {
                             return EXC7200_I2C_CoordRead_2ndTouch_Down;
                     }
                     if ( (Byte[1]==0x86))
                     {
                             return EXC7200_I2C_CoordRead_2ndtTouch_Lift;
                     }

   }


   return EXC7200_I2C_CoordRead_error;

}

// Example Interrupt function
void EXC7200_I2C_EXT_INT (void)
{

   u16 DataBuffer[10];
   u32 TPX1,TPY1,TPX2,TPY2;
   u16 Temp_X=0xFFFF,Temp_Y=0xFFFF;
   u16 temp;
   u8 Touch_size=4;
   u8    RStatus;

      while((ReadINT1())==0)
      {
                                    RStatus = EXC7200_I2C_CoordRead(0x04,10,DataBuffer);


                                    TPX1=(u16) DataBuffer[0]; //first X position
                                    TPY1=(u16) DataBuffer[1]; //first Y position
                                    TPX2=(u16) DataBuffer[3]; //second point X position
                                    TPY2=(u16) DataBuffer[4]; //second point Y position


                                     // Remapping Touch X,Y to LCD X,Y
                                    TPX1*=Current_LCM_ID.LCD_X_Max;
                                     TPX1/=2048;
                                    TPY1*=Current_LCM_ID.LCD_Y_Max;
                                    TPY1/=2048;
                                    TPX2*=Current_LCM_ID.LCD_X_Max;
                                     TPX2/=2048;
                                    TPY2*=Current_LCM_ID.LCD_Y_Max;
                                    TPY2/=2048;


                                  switch (RStatus)
                                  {
                                          case (EXC7200_I2C_CoordRead_1stTouch_Down):
                                  GUI_CircleFill(TPX1, TPY1, 2, rand()%0xFFFF);
```

```
                    // Do 1st touch down Function
                    break;
            case (EXC7200_I2C_CoordRead_1stTouch_Lift):
        GUI_RectangleFill(TPX1-4, TPY1-4,TPX1+4, TPY1+4 ,rand()%0xFFFF);
                    // Do 1st touch Lift Function

                    break;
            case (EXC7200_I2C_CoordRead_2ndTouch_Down):
        GUI_CircleFill(TPX1, TPY1, 2, RGB(128,128,128));
        GUI_CircleFill(TPX2, TPY2, 2, RGB(128,128,0));
                    // Do 2nd touch Down Function
                    break;
            case (EXC7200_I2C_CoordRead_2ndtTouch_Lift):
        GUI_RectangleFill(TPX1-4, TPY1-4,TPX1+4, TPY1+4 ,RGB(128,128,128));
        GUI_RectangleFill(TPX2-4, TPY2-4,TPX2+4, TPY2+4 ,RGB(128,128,0));
                    // Do 2nd touch Lift Function
                    break;
            default:
                    break;
        }

    }

}
```

9 DISPLAYED COLOR AND INPUT DATA

	Color & Gray Scale	DATA SIGNAL																	
		R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
Basic Color	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red(63)	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	Green(63)	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0
	Blue(63)	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1
	Cyan	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
	Magenta	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Red	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red(1)	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	Red(2)	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Red(31)	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Red(62)	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red(63)	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Green	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Green(1)	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
	Green(2)	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Green(31)	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Green(62)	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
	Green(63)	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0
Blue	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Blue(1)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	Blue(2)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Blue(31)	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	Blue(62)	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0
	Blue(63)	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1

10. RELIABILITY TEST CONDITIONS

Test Item	Test Conditions	Note
High Temperature Operation	70±3°C ,Dry t=240 hrs	
Low Temperature Operation	-20±3°C, Dry t=240 hrs	
High Temperature Storage	80±3°C , Dry t=240 hrs	1,2
Low Temperature Storage	-30±3°C ,Dry t=240 hrs	1,2
Thermal Shock Test	-20°C ~ 25°C ~ 70°C 30 m in. 5 min. 30 min. (1 cycle) Total 50cycle(Dry)	1,2
Storage Humidity Test	60 °C, Humidity 90%, 240 hrs	1,2
Vibration Test (Packing)	Sweep frequency : 10 ~ 55 ~ 10 Hz/1min Amplitude : 0.75mm Test direction : X.Y.Z/3 axis Duration : 30min/each axis	2

Note 1 : Condensation of water is not permitted on the module.

Note 2 : The module should be inspected after 1 hour storage in normal conditions

(15-35°C , 45-65%RH).

Definitions of life end point :

- Current drain should be smaller than the specific value.
- Function of the module should be maintained.
- Appearance and display quality should not have degraded noticeably.
- Contrast ratio should be greater than 50% of the initial value.

11. USE PRECAUTIONS

11-1 Handling precautions

(1) The polarizing plate may break easily so be careful when handling it. Do not touch, press or rub it with a hard-material tool like tweezers.

(2) Do not touch the polarizing plate surface with bare hands so as not to make it dirty. If the surface or other related part of the polarizing plate is dirty, soak a soft cotton cloth or chamois leather in benzine and wipe off with it. Do not use chemical liquids such as acetone, toluene and isopropyl alcohol. Failure to do so may bring chemical reaction phenomena and deteriorations.

(3) Remove any spit or water immediately. If it is left for hours, the suffered part may deform or decolorize.

(1) If the LCD element breaks and any LC stuff leaks, do not suck or lick it. Also if LC stuff is stuck on your skin or clothing, wash thoroughly with soap and water immediately.

11-2 Installing precautions

(1) The PCB has many ICs that may be damaged easily by static electricity. To prevent breaking by static electricity from the human body and clothing, earth the human body properly using the high resistance and discharge static electricity during the operation. In this case, however, the resistance value should be approx. 1MΩ and the resistance should be placed near the human body rather than the ground surface. When the indoor space is dry, static electricity may occur easily so be careful. We recommend the indoor space should be kept with humidity of 60% or more. When a soldering iron or other similar tool is used for assembly, be sure to earth it.

(2) When installing the module and ICs, do not bend or twist them. Failure to do so may crack LC element and cause circuit failure.

(3) To protect LC element, especially polarizing plate, use a transparent protective plate (e.g., acrylic plate, glass etc) for the product case.

(4) Do not use an adhesive like a both-side adhesive tape to make LCD surface (polarizing plate) and product case stick together. Failure to do so may cause the polarizing plate to peel off

11-3 Storage precautions

(1) Avoid a high temperature and humidity area. Keep the temperature between 0°C and 35°C and also the humidity under 60%.

(2) Choose the dark spaces where the product is not exposed to direct sunlight or fluorescent light.

(3) Store the products as they are put in the boxes provided from us or in the same conditions as we recommend.

11-4 Operating precautions

(1) Do not boost the applied drive voltage abnormally. Failure to do so may break ICs. When applying power voltage, check the electrical features beforehand and be careful. Always turn off the power to the LC module controller before removing or inserting the LC module input connector. If the input connector is removed or inserted while the power is turned on, the LC module internal circuit may break.

(2) The display response may be late if the operating temperature is under the normal standard, and the display may be out of order if it is above the normal standard. But this is not a failure; this will be restored if it is within the normal standard.

(3) The LCD contrast varies depending on the visual angle, ambient temperature, power voltage etc. Obtain the optimum contrast by adjusting the LC dive voltage.

(4) When carrying out the test, do not take the module out of the low-temperature space suddenly. Failure to do so will cause the module condensing, leading to malfunctions.

(5) Make certain that each signal noise level is within the standard (L level: 0.2Vdd or less and H level: 0.8Vdd or more) even if the module has functioned properly. If it is beyond the standard, the module may often malfunction. In addition, always connect the module when making noise level measurements.

(6) The CMOS ICs are incorporated in the module and the pull-up and pull-down function is not adopted for the input so avoid putting the input signal open while the power is ON.

(7) The characteristic of the semiconductor element changes when it is exposed to light emissions, therefore ICs on the LCD may malfunction if they receive light emissions. To prevent these malfunctions, design and assemble ICs so that they are shielded from light emissions.

(8) Crosstalk occurs because of characteristics of the LCD. In general, crosstalk occurs when the regularized display is maintained. Also, crosstalk is affected by the LC drive voltage. Design the contents of the display, considering crosstalk.

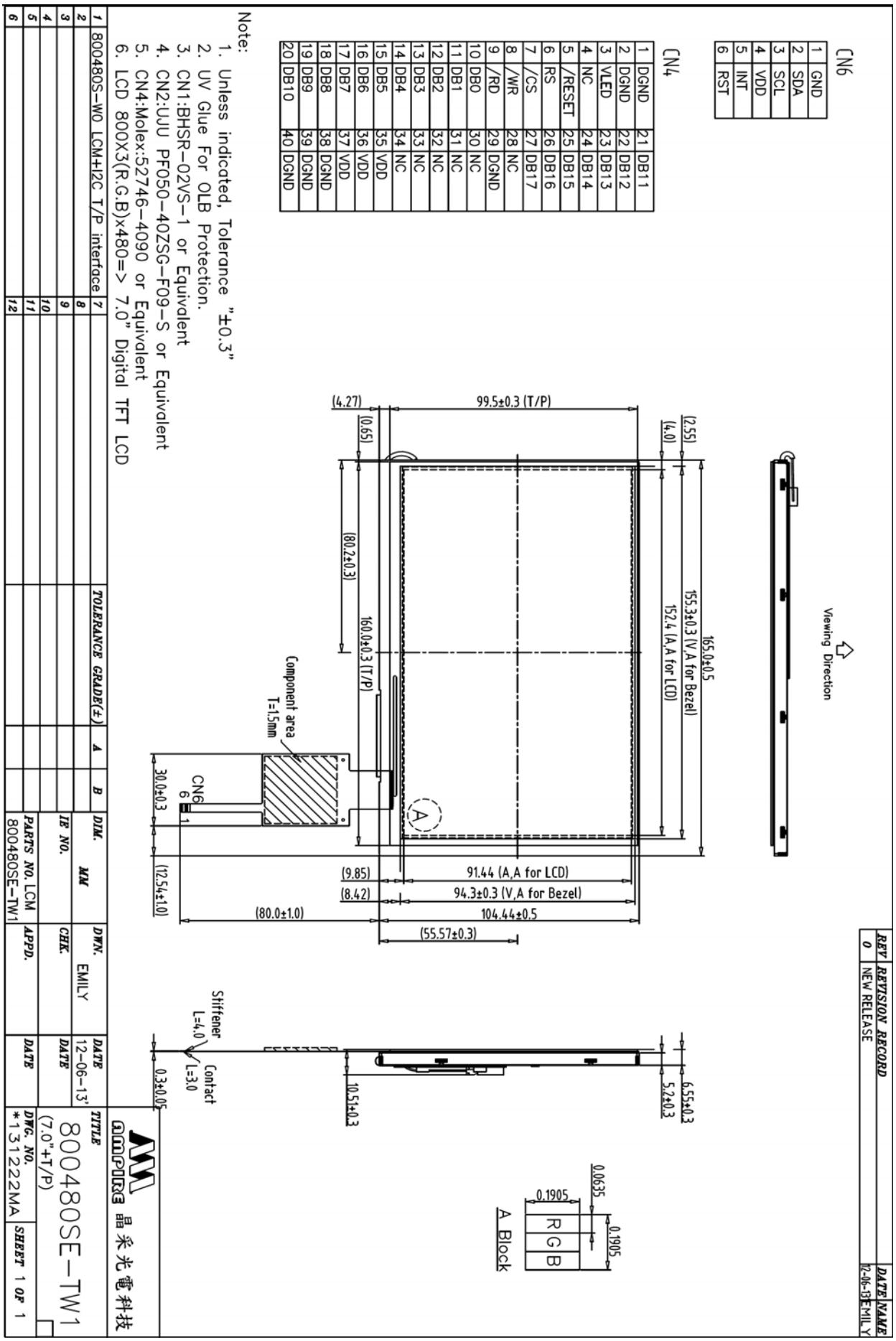
11-5 Other

(1) Do not disassemble or take the LC module into pieces. The LC modules once disassembled or taken into pieces are not the guarantee articles.

(2) The residual image may exist if the same display pattern is shown for hours. This residual image, however, disappears when another display pattern is shown or the drive is interrupted and left for a while. But this is not a problem on reliability.

(3) AMIPRE will provide one year warrantee for all products and three months warrantee for all repairing products..

12. OUTLINE DIMENSION



REV	REVISION RECORD	DATE	NAME
0	NEW RELEASE	12-06-13	EMILY

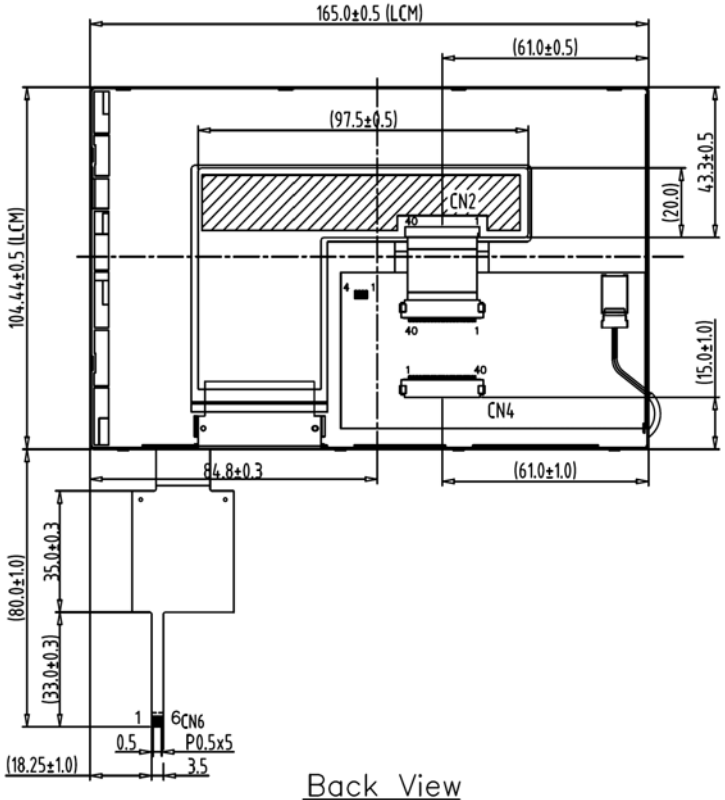
CN6

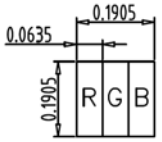
1	GND
2	SDA
3	SCL
4	VDD
5	INT
6	RST

CN4

1	DGND	21	DB11
2	DGND	22	DB12
3	VLED	23	DB13
4	NC	24	DB14
5	/RESET	25	DB15
6	RS	26	DB16
7	/CS	27	DB17
8	/WR	28	NC
9	/RD	29	DGND
10	DB0	30	NC
11	DB1	31	NC
12	DB2	32	NC
13	DB3	33	NC
14	DB4	34	NC
15	DB5	35	VDD
16	DB6	36	VDD
17	DB7	37	VDD
18	DB8	38	DGND
19	DB9	39	DGND
20	DB10	40	DGND

Note:
1. Unless indicated, Tolerance "±0.3"
2. UV Glue For OLB Protection.
3. CN1:BHSR-02VS-1 or Equivalent
4. CN2:UJU PF050-40ZSG-F09-S or Equivalent
5. CN4:Molex:52746-4090 or Equivalent
6. LCD 800X3(R.G.B)x480=> 7.0" Digital TFT LCD





A Block

1	800480S-W0 LCM+I2C T/P interface	7		TOLERANCE GRADE(±)	A	B	DIM. MM	DWN. EMILY	DATE 12-06-13'	TITLE 800480SE-TW1 (7.0"+T/P)	
2		8									
3		9					IE NO.	CHK.	DATE		
4		10									
5		11					PARTS NO. LCM-1 800480SE-TW1	APPD.	DATE	DWG. NO. *131223MA	SHEET 1 OF 1
6		12									

AMPIRE 晶采光電科技